



R2_DEPI2_CAI2_ISS4_S1_Cisco Network Administrator

Spring semester 2024-2025

Project Planning & Management for
Developing a Smart Office Network Solution

Supervisor

Dr.Rehab Mustafa

Submitted by: -

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Network Engineering Students Group code: CAI2_ISS4_S1

INTRODUCTION

Scenario

a smart office for a mid-sized business with about 100 employees. The office features IoT devices such as smart lighting, security cameras, and environmental sensors.

Requirements: The network must support various IoT devices, provide reliable internet access, ensure security for sensitive data, and enable remote monitoring and control of smart devices.

Objective

To design and implement a smart office network solution that meets the business's needs by:

- Supporting various IoT devices.
- Providing reliable and high-speed internet access.
- Ensuring security for sensitive business data.
- Enabling remote monitoring and control of smart devices.

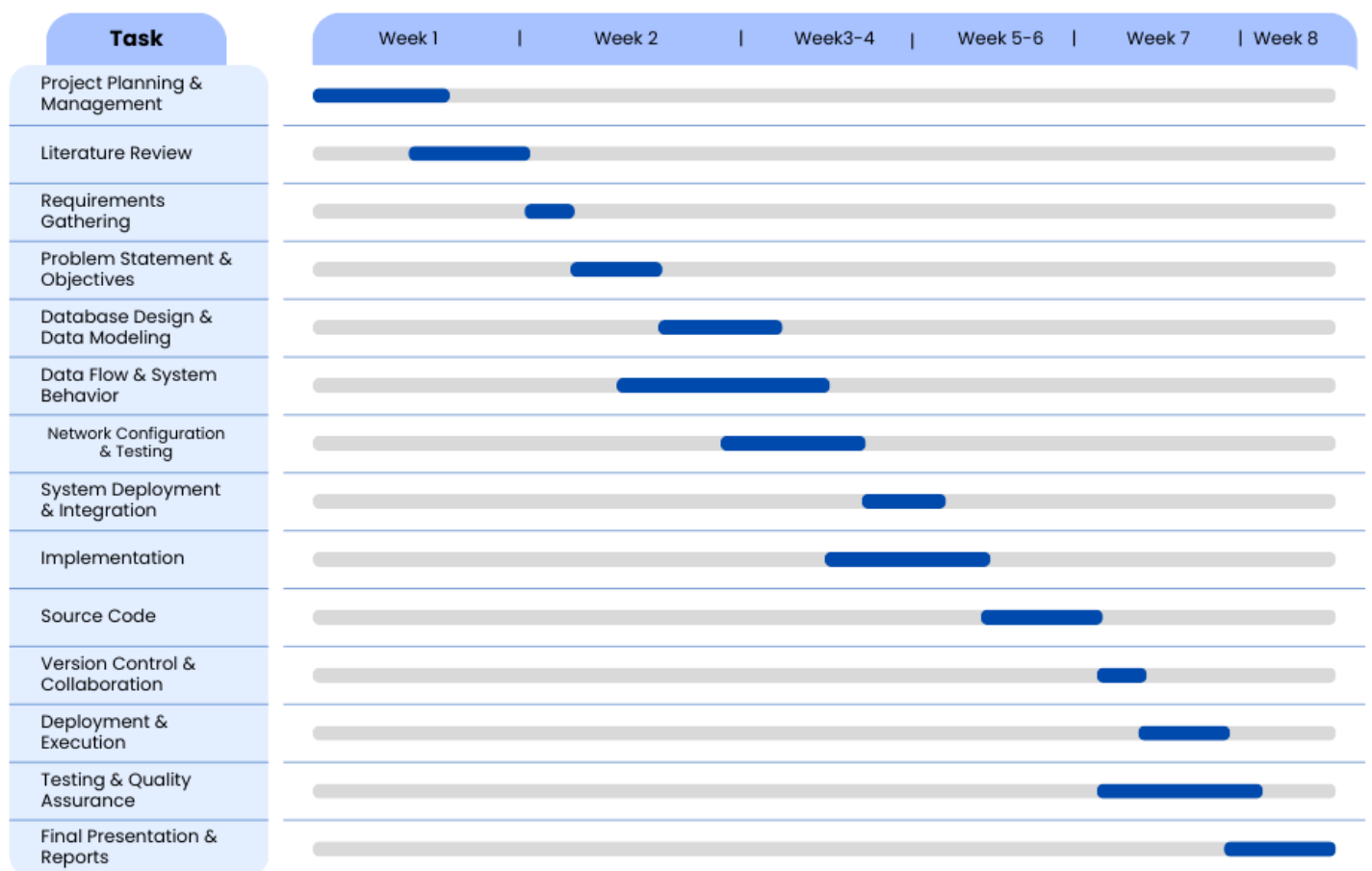
Scope

The project will cover:

- Network Infrastructure Design – Planning and implementing a scalable network.

- IoT Integration – Connecting and managing smart devices securely.
- Cybersecurity Measures – Implementing firewalls, encryption, and access controls.
- Remote Access & Monitoring – Enabling authorized users to monitor and control devices remotely.
- Scalability & Future Expansion – Ensuring the network can handle additional devices and users.

Project Timeline: Gantt Chart

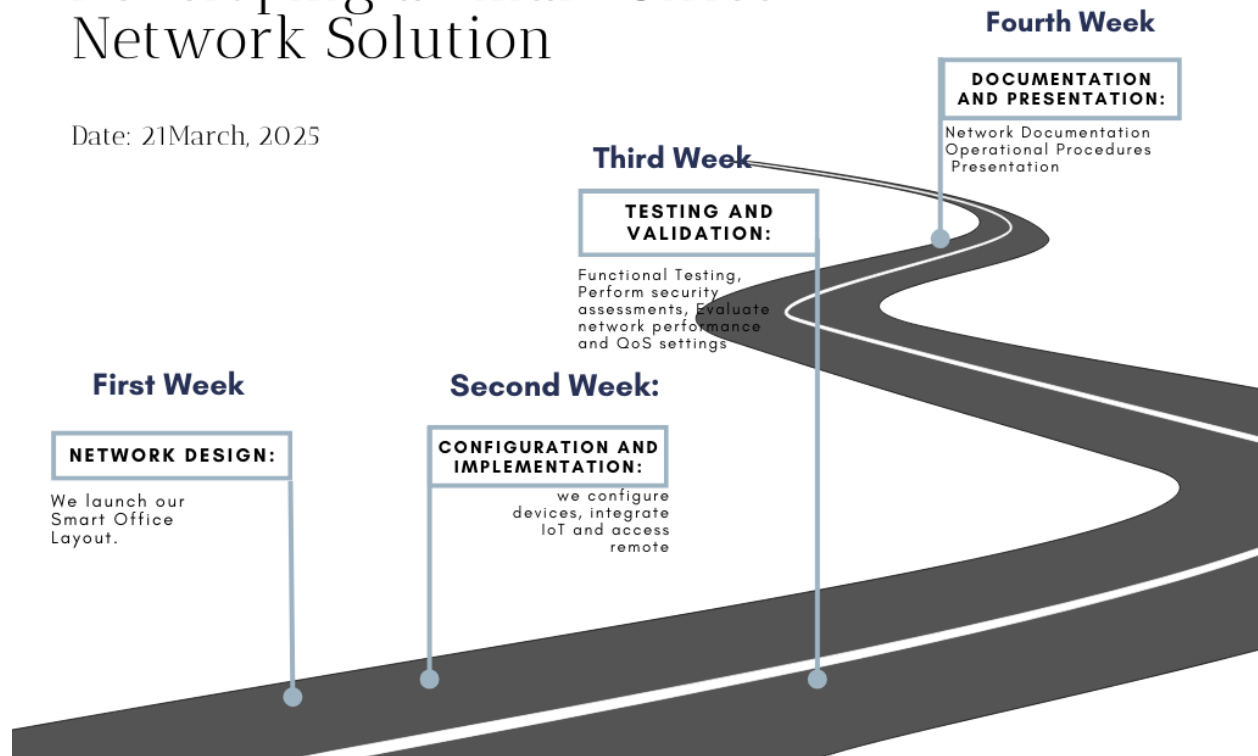


Milestones

PROJECT 14

Developing a Smart Office Network Solution

Date: 21March, 2025



Project Deliverable Acceptance Form

Project Name	Developing a Smart Office Network Solution
Project Manager	Dr. Rehab mustafa

These deliverables ensure the successful implementation of a secure, scalable, and efficient smart office network solution. Each phase of the project contributes to building a robust infrastructure that integrates IoT devices, maintains high network performance, and ensures data security. The documentation and training will facilitate smooth operations and future scalability, ensuring long-term sustainability of the network.

Deliverable	phase	Description
Project Planning & Management	Project Scope & Requirements Document	Establish the project scope, objectives, and initial timeline. Define key stakeholders and project requirements.
Network Design & Architecture	Network Design Document	Develop a detailed network design, including topology, IP addressing, security measures, and device configurations.
Hardware & Software Procurement	Procurement List	Identify and acquire necessary networking hardware, IoT devices, and software solutions required for implementation.
Network Implementation	Configured Network Infrastructure	Deploy and configure network components, integrate IoT devices, and ensure connectivity and functionality.
Security & Monitoring Setup	Security & Monitoring System	Implement cybersecurity measures, firewall configurations, and monitoring tools to ensure network security and reliability.
Testing & Optimization	Testing & Optimization Report	Conduct network performance testing, troubleshoot issues, and optimize configurations for efficiency and scalability.
Final Documentation & Handover	Final Report & User Training	Prepare project documentation, user manuals, and conduct training sessions for employees on network usage and security best practices.

Role	Responsibility	Name	Email
Employee NetworkManager	Set up and manage the main office network for employees	Mai Emad-Eldeen Mesllam Youssef Omar Hemdan	
IoT Network Manager	Configure and manage IoT devices (smart lighting, security cameras, sensors)	Manal Nabil Donia Mohamed saied Mustafa	
Guest Network Manager	Set up a secure guest network with restricted access to internal business resources.	Mohamed Hamza Ahmed	